

# Structural Radiocarbon Identification

## A Bayesian, Inverse-Geometric, and Multidisciplinary Framework for Improved Carbon Dating

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### Abstract

Radiocarbon dating is often presented as an exponential-decay inversion problem: an observed radiocarbon abundance is related through a decay law to an estimated age. A more general treatment regards dating as a Bayesian inverse-measurement problem in which isotope abundance is one component of an observation generated by a physical sample, a chemical preparation process, an instrument, a calibration system, and multiple sources of uncertainty. This treatise develops such a framework by combining Bayesian instrumentation, stochastic signal extraction, inverse statistical geometry, causal identification, constrained optimization, and adaptive experimental design.

The central principle is that, within a Bayesian measurement framework, the scientifically relevant output is a posterior distribution over latent states rather than necessarily a single numerical estimate. Structural constraints can then be used to distinguish ordinary measurement uncertainty from non-identification. The resulting framework preserves genuine ambiguity rather than converting it into artificial precision. Radiocarbon dating becomes a hierarchical identification problem involving decay physics, sample formation, reservoir effects, contamination, calibration, laboratory effects, and observation noise.

The framework is deliberately interdisciplinary. Physics supplies the decay process; chemistry supplies sample preparation and isotopic measurement; biology supplies carbon-cycle and organism-level formation mechanisms; statistics supplies likelihoods and uncertainty quantification; econometrics supplies identification and causal reasoning; information theory supplies uncertainty-reduction criteria; computer science supplies computational inference; economics supplies constrained resource allocation; and philosophy of science supplies the distinction between observation, model, and inference. Political science, geopolitics, diplomacy, welfare economics, and warfare economics are used only as conceptual analogies for measurement, information, incentives, and resource allocation, not as empirical evidence concerning radiocarbon dating.

The paper ends with “The End”

## 1 Introduction

The idealized radioactive-decay model for radiocarbon is

$$N_{14}(t) = N_{14,0} \exp(-\lambda t), \quad (1)$$

where  $N_{14,0}$  is an initial amount and  $\lambda$  is a decay constant.

If both  $N_{14,0}$  and the relevant measured quantity were known without uncertainty, the inverse relationship would be

$$t = -\frac{1}{\lambda} \log \left( \frac{N_{14}(t)}{N_{14,0}} \right). \quad (2)$$

The mathematical inversion is therefore simple. The scientific problem is the construction of a defensible relationship between the measured quantity and the latent historical state represented by  $t$ .

The General Bayesian Theory of Instrumentation develops a framework in which an instrument is treated as a Bayesian dynamical system and a measurement is represented by posterior inference rather than merely by a point estimate [1]. The General Theory of Signal Extraction begins from a generic observation equation separating latent signal, carrier or measurement mechanism, and noise [2]. Inverse Statistical Geometry and Bayesian Causal Inference add the idea that observations may admit multiple latent realizations and that physical or causal restrictions can reduce this ambiguity [3, 4, 6].

These ideas are directly relevant to radiocarbon dating.  
The central question is therefore:

Given an imperfect observation generated by a historical physical system, what can be identified about the latent age under the available physical, chemical, biological, instrumental, and statistical constraints?

The proposed answer is to construct the full generative system, infer its latent states jointly, impose scientifically justified restrictions, explicitly represent structural non-identification, and allocate additional measurement resources according to a stated information or decision criterion.

## 2 Measurement as Bayesian Inference

Let the complete latent state be

$$\theta = (\tau, N_{14,0}, \lambda, \eta, \Theta, C, X), \quad (3)$$

where  $\tau$  is target age,  $\eta$  denotes sample-specific nuisance parameters,  $\Theta$  denotes instrument and carrier parameters,  $C$  denotes calibration parameters or calibration uncertainty, and  $X$  denotes latent physical or chemical states.

Given observations  $Y$ , Bayesian inference gives

$$p(\theta | Y) = \frac{p(Y | \theta)p(\theta)}{\int p(Y | \theta)p(\theta), d\theta}. \quad (4)$$

The marginal posterior for age is

$$p(\tau | Y) = \int p(\theta | Y), d\theta_{-\tau}, \quad (5)$$

where  $d\theta_{-\tau}$  denotes integration over all latent components other than  $\tau$ .

Thus, within the Bayesian instrumentation framework, the age measurement is a distribution over admissible ages rather than necessarily a single value [1].

## 3 A Layered Observation Model

A radiocarbon experiment can be represented as a sequence

historical process  $\rightarrow$  carbon state  $\rightarrow$  sample  $\rightarrow$  chemical preparation  $\rightarrow$  instrument  $\rightarrow$  data  $\rightarrow$  posterior. (6)

A general stochastic state-space representation is

$$dX_t = f(X_t, \theta, t), dt + G(X_t, \theta, t), dW_t, \quad (7)$$

with observations

$$dY_t = h(X_t, \theta, t), dt + R^{1/2}, dV_t. \quad (8)$$

Here  $W_t$  and  $V_t$  represent process and observation stochasticity.

This representation need not imply that every radiocarbon laboratory experiment should literally be implemented as a continuous-time stochastic differential equation. Rather, it provides a general language for separating latent physical states from observations [2, 7].

## 4 Radiocarbon Dating as an Inverse Problem

Suppose that

$$Y = S + \epsilon, \quad \epsilon \sim N(0, \sigma^2 I). \quad (9)$$

The inverse problem asks which latent signal  $S$  could have generated  $Y$ . Bayesian inference gives

$$p(S, G | Y) \propto p(Y | S, G)p(S, G). \quad (10)$$

For radiocarbon dating,  $S$  can represent a latent physical and chemical carbon state while  $Y$  can include isotope measurements, laboratory measurements, standards, blanks, replicates, and auxiliary calibration information.

The crucial difficulty is non-uniqueness. Define the realization set

$$M_R(Y) = \{\theta : R(\theta) = R(Y)\}, \quad (11)$$

where  $R$  denotes a selected relational or structural representation.

This set contains latent parameter states observationally equivalent under  $R$ . In general,

$$|M_R(Y)| > 1. \quad (12)$$

Consequently, numerical precision in an estimator does not by itself establish structural identification.

## 5 Physical and Scientific Constraints

Let  $M_P$  denote the set of parameter states compatible with the applicable physical, chemical, biological, instrumental, sampling, and calibration constraints.

The admissible state space is

$$M^*(Y) = M_R(Y) \cap M_P. \quad (13)$$

If

$$M^*(Y) = \theta^*, \quad (14)$$

then the relevant parameter state is structurally identified relative to the specified model and constraints.

If

$$|M^*(Y)| > 1, \quad (15)$$

then multiple admissible explanations remain.

A constrained posterior can therefore be written

$$p(\theta | Y, R, M_P) \propto p(Y | \theta)p(\theta)\mathbf{1}_{\theta \in M_R(Y) \cap M_P}. \quad (16)$$

The framework therefore treats identification as a property of the observation model together with the imposed scientific restrictions, rather than as a property of numerical estimation alone [3, 4].

## 6 Posterior Identification and Consistency

The original formulation of the identification argument requires an important qualification. A singleton admissible set does not, by itself, mathematically guarantee posterior convergence. Posterior consistency also requires appropriate regularity, correct specification, sufficient information, and a prior that does not exclude neighborhoods of the true state.

A more precise statement is therefore the following.

### Proposition: Structural Posterior Consistency

Let  $\theta^*$  be the true parameter state. Suppose that:

1. the statistical model is correctly specified;
2. the admissible parameter space is  $M_P$ ;
3.  $\theta^*$  is uniquely identified in  $M_P$ ;
4. the prior assigns positive probability to every sufficiently small neighborhood of  $\theta^*$ ;
5. the likelihood satisfies the regularity and separation conditions required for posterior consistency.

Then, for observations  $Y_1, \dots, Y_n$  whose information increases with  $n$ ,

$$p(\theta \in U \mid Y_1, \dots, Y_n) \longrightarrow 1 \quad (17)$$

for every neighborhood  $U$  of  $\theta^*$ .

### Proof sketch

Unique identification implies that competing admissible parameter values cannot reproduce the data-generating distribution under the maintained model. Under the stated posterior-consistency conditions, likelihood accumulation separates neighborhoods of  $\theta^*$  from competing regions. The prior remains positive in neighborhoods of the true state. Consequently, posterior mass outside every neighborhood of  $\theta^*$  converges to zero.

For continuous parameters, the appropriate conclusion is concentration in arbitrarily small neighborhoods, not necessarily that a literal point mass appears at a finite or even asymptotic parameter value. This distinction is essential.

If instead

$$M_P \cap M_R(Y) = \theta_1^*, \dots, \theta_k^*, \quad k > 1, \quad (18)$$

then structural non-identification remains and should be represented by the posterior.

## 7 Calibration as a Latent Scientific Object

Calibration need not be represented as a deterministic transformation. Let  $C$  denote the calibration state or calibration function.

Then

$$p(\tau, C, \eta, \Theta \mid Y) \propto p(Y \mid \tau, C, \eta, \Theta) p(C) p(\tau) p(\eta) p(\Theta). \quad (19)$$

The age posterior is

$$p(\tau \mid Y) = \int p(\tau, C, \eta, \Theta \mid Y) dC d\eta d\Theta. \quad (20)$$

This separates calibration uncertainty from ordinary observation noise.

It also makes sensitivity analysis possible: alternative calibration specifications can be compared through posterior distributions rather than hidden inside a single deterministic preprocessing step.

## 8 Hierarchical Multi-Laboratory Measurement

Suppose laboratories  $i = 1, \dots, n$  provide observations  $D_i$ . A simple conditional-independence model is

$$p(\theta \mid D_1, \dots, D_n) \propto p(\theta) \prod_{i=1}^n p(D_i \mid \theta). \quad (21)$$

However, conditional independence may be inappropriate when laboratories share standards, calibration systems, preparation protocols, instruments, or other systematic components.

Introduce laboratory-specific effects  $b_i$  and a shared effect  $b_0$ :

$$D_i = h_i(\theta, b_0, b_i) + \epsilon_i. \quad (22)$$

The joint posterior becomes

$$p(\theta, b_0, b_1, \dots, b_n \mid D_1, \dots, D_n) \propto p(\theta)p(b_0) \prod_i p(b_i) \prod_i p(D_i \mid \theta, b_0, b_i). \quad (23)$$

This prevents replicated measurements from automatically being treated as independent information when they share systematic uncertainty.

## 9 Chemical and Biological Structure

The measured carbon reflects biological and chemical history. A generic carbon-pool model may be written

$$\frac{dX_j}{dt} = \sum_k F_{kj}(X, t) - \sum_k F_{jk}(X, t) - \lambda_j X_j, \quad (24)$$

where  $X_j$  denotes a carbon pool and  $F_{jk}$  denotes transfer between pools.

The target age is consequently not necessarily identical to the elapsed time since the carbon last exchanged with the atmosphere.

This motivates nuisance or structural parameters such as

$$\eta = (\eta_{\text{reservoir}}, \eta_{\text{contamination}}, \eta_{\text{exchange}}, \eta_{\text{preservation}}, \eta_{\text{preparation}}). \quad (25)$$

The functional form and prior distribution of these quantities must be established from the relevant scientific evidence. The present framework does not assert universal parameterizations for them.

## 10 Contamination as a Mixture Model

A simple contamination representation is

$$N_{\text{obs}} = (1 - \alpha)N_{\text{sample}} + \alpha N_{\text{cont}}, \quad 0 \leq \alpha \leq 1. \quad (26)$$

The contamination fraction  $\alpha$  can be treated as an unknown parameter with prior  $p(\alpha)$ :

$$p(\tau, \alpha \mid Y) \propto p(Y \mid \tau, \alpha)p(\tau)p(\alpha). \quad (27)$$

More complicated contamination processes can be represented by mixtures over multiple contaminant sources.

The methodological point is that contamination becomes an explicit uncertainty component rather than an informal caveat.

## 11 Information-Theoretic Experimental Design

Let  $u$  denote a possible additional experiment and  $c(u)$  its resource cost.

For a continuous age distribution, the quantity

$$H(\tau \mid D) = - \int p(\tau \mid D) \log p(\tau \mid D), d\tau \quad (28)$$

is differential entropy. It can be useful for comparing posterior uncertainty within a fixed parameterization, although it is not invariant under arbitrary reparameterization and should therefore not be treated as an absolute measure of uncertainty.

Expected information gain is more robustly expressed through Kullback–Leibler divergence. For prospective experiment  $u$ ,

$$\text{EIG}(u) = \mathbb{E} * D_u \mid D, u [D * \text{KL}(p(\tau \mid D, D_u, u), \parallel, p(\tau \mid D))]. \quad (29)$$

An information-per-cost criterion is then

$$\mathcal{E}(u) = \frac{\text{EIG}(u)}{c(u)}. \quad (30)$$

If the scientific objective is explicitly to minimize expected posterior entropy plus cost, an alternative decision criterion is

$$u^* = \arg \min_u [c(u) + \lambda \mathbb{E}[H(\tau \mid D, D_u, u)]] . \quad (31)$$

These are alternative objectives, not mathematically identical optimization criteria.

## 12 Economic Allocation of Laboratory Resources

Let  $x_i$  denote the intensity or number of measurements of type  $i$ , with cost  $c_i$ , and let  $B$  be the available budget.

An information-maximizing design may be written, for a specified information criterion  $\mathcal{I}$ ,

$$\max_{x_1, \dots, x_m} \mathcal{I}(x_1, \dots, x_m) \quad (32)$$

subject to

$$\sum_{i=1}^m c_i x_i \leq B, \quad (33)$$

$$x_i \geq 0. \quad (34)$$

The Lagrangian is

$$\mathcal{L} = \mathcal{I}(x) - \mu \left( \sum_i c_i x_i - B \right). \quad (35)$$

At an interior optimum,

$$\frac{\partial \mathcal{I}}{\partial x_i} = \mu c_i. \quad (36)$$

This is the familiar marginal-value condition from constrained optimization and provides an economic interpretation of experimental design [8].

## 13 Econometrics and Structural Identification

Consider

$$Y = g(\tau, \eta) + \epsilon. \quad (37)$$

Even if a statistical estimator is numerically well behaved,  $\tau$  may remain structurally unidentified if distinct parameter states generate observationally equivalent distributions.

For example,

$$p(Y \mid \tau_1, \eta_1) = p(Y \mid \tau_2, \eta_2) \quad (38)$$

for distinct parameter pairs implies that the observation model alone cannot distinguish those states. Define the observational equivalence class

$$\mathcal{E}(Y) = \{(\tau, \eta) : p(Y \mid \tau, \eta) = p(Y \mid \theta_{\text{obs}})\}. \quad (39)$$

Additional physical, chemical, biological, or causal restrictions can reduce this equivalence class.

Thus radiocarbon inference has the same fundamental logical distinction encountered in econometrics:

$$\text{estimation} \neq \text{identification}. \quad (40)$$

## 14 Causal Structure

A useful causal representation is

$$\text{historical environment} \rightarrow \text{carbon reservoir} \rightarrow \text{organism or material} \rightarrow \text{sample} \rightarrow \text{measurement}. \quad (41)$$

Each arrow represents a mechanism rather than a mere statistical association.

The Bayesian causal framework supplied in the source material combines stochastic dynamics, Bayesian inference, and causal restrictions [6].

The same principle can be applied here: scientifically justified causal restrictions can eliminate latent explanations that would otherwise remain statistically compatible with the observations.

## 15 Physics of Instrumentation

A generalized physical instrument can be represented by

$$M(x, t)\dot{x} + K(x, t)x = B(x, t)u, \quad (42)$$

with observation equation

$$D_t = H(x(t)) + \epsilon_t. \quad (43)$$

The corresponding Bayesian measurement is

$$p(\theta \mid D_t) = \frac{p(D_t \mid \theta)p(\theta)}{p(D_t)}. \quad (44)$$

This formalizes the idea that the instrument is a transformation between physical state and information state [1].

The RLC example in the supplied material demonstrates the broader use of stochastic differential equations and stochastic optimal control for physical measurement systems [7]. Its purpose here is methodological rather than radiocarbon-specific.

## 16 Signal Processing and Filtering

If a latent state evolves according to

$$dX_t = f(X_t, t)dt + Q^{1/2}dW_t, \quad (45)$$

with observation process

$$dY_t = h(X_t, t)dt + R^{1/2}dV_t, \quad (46)$$

the filtering target is

$$p(X_t \mid Y_{0:t}). \quad (47)$$

For linear-Gaussian systems, Kalman filtering is available. For nonlinear or non-Gaussian systems, particle filtering, sequential Monte Carlo, variational inference, or other methods may be appropriate [2].

The algorithmic sophistication, however, cannot substitute for a scientifically appropriate generative model.

## 17 Stochastic Control and Experimental Action

Let the state evolve according to

$$dX_t = f(X_t, u_t, t)dt + G(X_t, u_t, t)dW_t. \quad (48)$$

For a value function  $V(x, t)$ , a Hamilton–Jacobi–Bellman equation is

$$0 = V_t + \min_u \left\{ L(x, u, t) + \nabla V^\top f + \frac{1}{2} \text{tr} (GG^\top \nabla^2 V) \right\}. \quad (49)$$

In the radiocarbon application,  $u$  may represent an experimental action such as additional measurement, replication, chemical reprocessing, calibration measurement, or selection of another measurement channel.

The optimization target is therefore not ‘choose the age.’ It is ‘choose a scientific action under uncertainty.’

This is consistent with the stochastic-control framework developed in the supplied Ghoshian manuscripts [6, 7].

## 18 Precision, Uncertainty, and Human Interpretation

A central methodological distinction is

$$\text{precision} \neq \text{identification}. \quad (50)$$

A narrow posterior can arise because a model is highly restrictive. That does not by itself establish that the underlying historical state is uniquely identified.

A useful conceptual decomposition is

$$U_{\text{total}} = U_{\text{measurement}} + U_{\text{calibration}} + U_{\text{model}} + U_{\text{structural}} + U_{\text{sampling}}, \quad (51)$$

where the notation is a conceptual decomposition rather than an assumption that the components are statistically independent.

This distinction is important for scientific communication because a numerical interval can be interpreted as stronger evidence than warranted if model and structural uncertainty are hidden.

## 19 Philosophy of Science

The framework separates three propositions:

1. an observation was recorded;
2. a model explains the observation;
3. the model identifies a historical state.

These propositions are not logically equivalent.

In particular,

$$\text{good statistical fit} \not\Rightarrow \text{unique historical explanation}, \quad (52)$$

and

$$\text{small sampling uncertainty} \not\Rightarrow \text{small model uncertainty}. \quad (53)$$

The inverse-geometric perspective therefore treats scientific inference as constrained reasoning over possible latent states.

## 20 Religion, History, and Chronology

Religious traditions, historical narratives, and cultural memory may contain claims about chronology, ancestry, migration, continuity, creation, or destruction. Such narratives may motivate historical hypotheses, but they are not automatically quantitative radiocarbon observations.

Consequently,

$$\text{cultural narrative} \neq \text{instrumental observation}. \quad (54)$$

Nevertheless, a historically informed hypothesis can affect sampling strategy or prior specification when that use is explicitly justified.

The framework therefore distinguishes the epistemic roles of narrative evidence, auxiliary evidence, and instrumental measurements.

## 21 Political Science, Geopolitics, and Diplomacy

Political and international systems provide conceptual examples of a broader measurement problem: an observed statement, action, or signal need not equal the latent state that generated it.

The generic structure is

$$\text{latent state} \longrightarrow \text{institutional process} \longrightarrow \text{observable signal}. \quad (55)$$

This is analogous, at an abstract level, to

$$\text{historical carbon state} \rightarrow \text{sample and laboratory process} \rightarrow \text{observed isotope signal}. \quad (56)$$

The analogy is methodological only. Political or geopolitical considerations do not constitute empirical evidence for a radiocarbon age.

## 22 Welfare Economics and Decision Theory

Suppose a downstream decision  $a$  depends on the unknown age  $\tau$ . Let  $L(a, \tau)$  denote the loss associated with decision  $a$  when the true state is  $\tau$ .

The Bayesian decision rule is

$$a^* = \arg \min_a \mathbb{E}[L(a, \tau) \mid D]. \quad (57)$$

This separates uncertainty about the state from preferences over consequences.

Thus a scientific age estimate need not itself determine a decision. The posterior supplies information; the loss function represents the decision-maker's objective.

## 23 Warfare Economics as a General Resource-Allocation Analogy

Warfare economics provides an extreme example of resource allocation under uncertainty. A generic stochastic decision problem is

$$\max_u \mathbb{E}[B(u, \theta) \mid D] - C(u). \quad (58)$$

The analogous laboratory problem is to choose additional measurements according to expected scientific value and resource cost.

This analogy does not imply that radiocarbon laboratories should be organized according to military principles. It illustrates only that constrained stochastic optimization has mathematical structures common to many domains.

## 24 Computer Science and Computational Architecture

A practical Bayesian Structural Radiocarbon Dating system can be organized as

|                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{Data} \rightarrow \begin{matrix} \text{Physical} \\ \text{Model} \end{matrix} \rightarrow \begin{matrix} \text{Instrument} \\ \text{Model} \end{matrix} \rightarrow \begin{matrix} \text{Signal} \\ \text{Extraction} \end{matrix} \rightarrow \text{Identification} \rightarrow \text{Posterior} \rightarrow \text{Decision} \quad (59)$ |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

A corresponding computational procedure is:

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INPUT:
isotope observations
chemical observations
sample metadata
laboratory metadata
calibration information
physical constraints
resource constraints

INITIALIZE:
priors
physical state model
observation model
calibration model

REPEAT:
1. construct likelihood
2. update posterior
3. extract latent signal
4. construct realization set
5. impose physical and causal constraints
6. evaluate identification
7. propagate calibration uncertainty
8. perform posterior diagnostics
9. enumerate feasible additional experiments
10. calculate expected information gain
11. evaluate information relative to cost
12. select an additional experiment if justified
UNTIL:
posterior convergence OR
stopping criterion OR
resource constraint binds

OUTPUT:
posterior age distribution
uncertainty decomposition
identification status
sensitivity analysis
posterior predictive diagnostics
recommended additional measurements

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## 25 Artificial Intelligence and Machine Learning

Machine learning can assist with nonlinear signal extraction, anomaly detection, quality control, surrogate likelihoods, posterior approximation, experimental design, and simulation-based calibration.

A generic amortized posterior approximation may be written

$$q_{\phi}(\theta \mid Y) \approx p(\theta \mid Y), \quad (60)$$

where  $\phi$  denotes learned parameters.

An optimization objective may be

$$\hat{\phi} = \arg \min_{\phi} \mathbb{E} [\mathcal{L}(q_{\phi}(\theta \mid Y), p(\theta \mid Y))]. \quad (61)$$

However,

$$\text{predictive accuracy} \neq \text{causal identification} \neq \text{physical validity}. \quad (62)$$

A flexible machine-learning model can approximate a conditional distribution without proving that the latent historical state is uniquely identified.

## 26 Simulation and Empirical Validation

The proposed framework should not be declared superior merely because it is more elaborate. It requires empirical validation.

Synthetic samples with known latent ages can be generated according to

$$\tau_j^{\text{true}} \sim p(\tau), \quad (63)$$

followed by

$$Y_j \sim p(Y \mid \tau_j^{\text{true}}, \eta_j, \Theta_j, C). \quad (64)$$

Conventional and structural Bayesian methods can then be compared.

Table 1: Proposed validation metrics

| Metric                 | Purpose                                  |
|------------------------|------------------------------------------|
| Absolute error         | Point-estimation accuracy                |
| Posterior coverage     | Calibration of credible intervals        |
| Log predictive density | Probabilistic predictive quality         |
| CRPS                   | Distributional forecast accuracy         |
| Posterior entropy      | Within-model uncertainty assessment      |
| Identification rate    | Frequency of unique admissible solutions |
| Sensitivity            | Dependence on modeling assumptions       |
| Robustness             | Stability under perturbations            |
| Information per cost   | Experimental efficiency                  |

The central empirical comparison is not whether the proposed method produces narrower intervals. It is whether those intervals remain calibrated and whether structural uncertainty is represented correctly.

## 27 Structural Stress Testing

The framework should be subjected to deliberate perturbations:

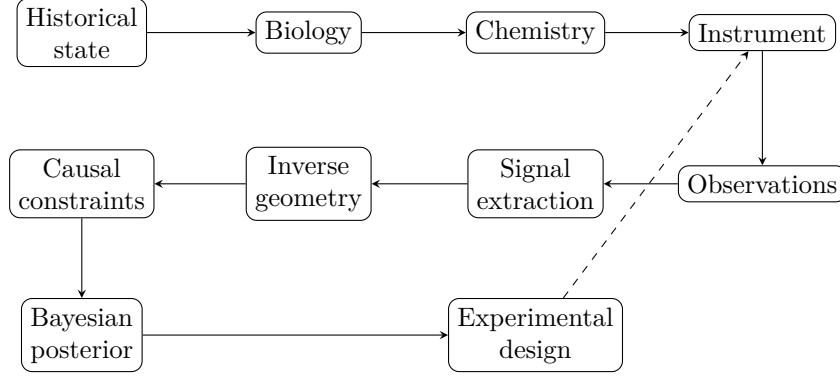
1. calibration misspecification;
2. contamination;
3. reservoir effects;
4. correlated laboratory errors;
5. misspecified decay parameters;
6. nonstationary environmental processes;
7. unmodeled sample mixtures;
8. sparse observations;
9. extreme-age observations;
10. contradictory auxiliary information.

For perturbation  $\delta$ , define

$$\Delta_\delta = d(p(\tau \mid Y), p_\delta(\tau \mid Y)), \quad (65)$$

where  $d$  is a selected probability-distance or discrepancy measure.

A robust methodology should expose sensitivity rather than conceal it.



## 28 Bayesian Model Comparison

Suppose competing models are  $M_1, \dots, M_K$ . Their posterior probabilities are

$$p(M_k | Y) \propto p(Y | M_k)p(M_k). \quad (66)$$

The model-averaged age posterior is

$$p(\tau | Y) = \sum_{k=1}^K p(\tau | Y, M_k)p(M_k | Y). \quad (67)$$

Model averaging provides a principled mechanism for propagating model uncertainty into the final age distribution.

It also makes explicit the fact that a selected model is itself an inferential object.

## 29 System Architecture

The dashed arrow represents adaptive experimental design: posterior information is used to select a subsequent measurement.

## 30 What the Framework Changes

The conventional conceptual chain is approximately

$$\text{radiocarbon measurement} \rightarrow \text{calibrated age}. \quad (68)$$

The proposed chain is

$$\begin{aligned} \text{historical state} &\rightarrow \text{physical process} \rightarrow \text{biological and chemical transformation} \rightarrow \text{sample} \rightarrow \text{instrument} \rightarrow \text{observed data} \\ &\hspace{15em} \rightarrow \text{signal extraction} \rightarrow \text{structural identification} \rightarrow \text{posterior age distribution} . \end{aligned} \quad (69)$$

The second representation is more explicit about assumptions and uncertainty.

Its methodological advantages are therefore:

1. uncertainty can be decomposed rather than hidden inside one standard error;
2. calibration uncertainty can be represented explicitly;
3. contamination can be modeled as a latent quantity;
4. correlated laboratory effects can be represented;
5. physical and causal restrictions can contribute to identification;
6. structural ambiguity can be preserved;

7. additional measurements can be selected according to an explicit criterion;
8. model uncertainty can be propagated;
9. computational methods can be evaluated using calibration, coverage, predictive performance, and robustness rather than apparent precision alone.

## 31 Limits and Empirical Status

The framework is a methodological proposal, not an empirical demonstration that existing radiocarbon laboratories are systematically incorrect.

The supplied source materials establish general theories of Bayesian instrumentation, signal extraction, inverse statistical geometry, causal inference, stochastic control, and resource allocation [1, 2, 3, 6, 8]. They do not themselves constitute radiocarbon-specific laboratory validation.

Consequently,

$$\text{methodological extension} \neq \text{empirical validation}. \quad (70)$$

Radiocarbon-specific claims require appropriate archaeological, geological, chemical, biological, and laboratory datasets.

The framework should therefore be evaluated using synthetic benchmarks, known-age materials, interlaboratory comparisons, replicated measurements, and prospective validation.

## 32 Conclusion

Radiocarbon dating can be reformulated as a Bayesian inverse-measurement and structural-identification problem.

The central object is not merely a number produced by an instrument. It is a posterior distribution over latent historical states conditional on observations, scientific constraints, calibration information, and model assumptions:

$$p(\tau | Y) = \int p(\tau, \eta, \Theta, C, X | Y), d\eta, d\Theta, dC, dX. \quad (71)$$

The posterior is constrained by

$$\theta \in M_R(Y) \cap M_P. \quad (72)$$

If the constrained model is identifiable and satisfies appropriate posterior-consistency conditions, additional information can concentrate the posterior around the true state. If multiple scientifically admissible explanations remain, the posterior should preserve that ambiguity.

The resulting methodological principle is

$$\boxed{\text{measurement} + \frac{\text{structural}}{\text{constraints}} + \frac{\text{uncertainty}}{\text{quantification}} + \frac{\text{adaptive}}{\text{information acquisition}} = \frac{\text{improved}}{\text{scientific identification}}} \quad (73)$$

This principle extends beyond radiocarbon dating. Physics, chemistry, biology, econometrics, statistics, computer science, economics, information theory, causal inference, and decision theory all encounter variants of the same problem:

$$\boxed{\text{What can be learned about an unobserved state from an imperfect observation?}} \quad (74)$$

The answer depends not only on the amount of data, but also on the geometry of the admissible state space, the validity of the measurement model, the strength of physical and causal restrictions, the structure of uncertainty, and the allocation of scarce information-gathering resources.

The principal improvement proposed here is therefore not a new decay equation. It is a more explicit architecture for scientific inference: carbon dating becomes a system of measurement, identification, uncertainty quantification, model comparison, and adaptive experimentation.

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## Glossary

### Bayesian Instrumentation

A framework in which an instrument and its observations are represented as a probabilistic inference system mapping observations to posterior distributions over latent states.

### Calibration

A scientifically justified relationship connecting an instrumental or measured quantity to a target quantity, together with uncertainty in that relationship.

### Causal Identification

Determination of a latent causal quantity from observations under structural assumptions that distinguish otherwise observationally equivalent explanations.

### Contamination

Addition of carbon or other material not belonging to the target sample, potentially changing the observed isotopic composition.

### Experimental Design

Selection of observations or interventions to achieve a stated scientific objective subject to resource constraints.

### Inverse Statistical Geometry

The study of recovering latent realizations from statistical or relational structures generated by observations.

### Model Uncertainty

Uncertainty concerning which mathematical, statistical, chemical, biological, or physical model adequately represents the data-generating process.

**Posterior Distribution**

The probability distribution over latent parameters after combining prior information with observed data.

**Posterior Consistency**

Concentration of posterior probability around the true parameter as informative observations accumulate, subject to appropriate regularity conditions.

**Realization Set**

The collection of latent parameter states producing the same specified observational or relational structure.

**Signal Extraction**

Recovery of a latent information-bearing state from noisy observations.

**State-Space Model**

A model separating latent dynamical states from their observations.

**Structural Ambiguity**

Persistence of multiple scientifically admissible latent explanations after conditioning on the available observations and constraints.

**Structural Identification**

Unique determination of the relevant latent state within a specified model and admissible parameter space.

**Uncertainty Decomposition**

Explicit separation of conceptually distinct uncertainty sources such as measurement, calibration, sampling, model, and structural uncertainty.

**The End**